Cyclic Quotient Systems in Empirical Economics

A G20 Panel Analysis of Inequality, Population Density, and Income

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Abstract

This paper presents an empirical illustration of cyclic quotient systems (CQS) using a country-year panel of 19 sovereign G20 members over 2000–2023. The repository's saved data contain 456 country-year rows, of which 327 have an observed World Bank Gini index; the complete-case regression sample includes 327 observations from 18 countries over 24 years. The paper distinguishes the exact algebra of quotient closure from empirical evidence. The population-area-Gini product is one, up to floating-point rounding, because its three ratios are constructed from the same country-year values. This is a coding consistency check, not an empirical finding or noisy-cycle test. Separately, a descriptive two-way fixed-effects regression relates log Gini to log population density and log real GDP per capita. The estimated coefficients are 0.0186 for log density and 0.0475 for log GDP per capita, with country-clustered standard errors of 0.330 and 0.096, respectively. Neither coefficient is statistically distinguishable from zero at conventional levels. These associations are not causal estimates. Sparse and unbalanced Gini coverage, only 18 regression clusters, a high reported condition number, and low Durbin–Watson statistic warrant caution. The paper interprets the repository's stored results and sets out requirements for stronger inference and replication.

The paper ends with "The End"

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1 Introduction

A cyclic quotient system arranges ratios around a closed sequence of quantities. Let $x_1, \ldots, x_N$ be nonzero quantities and set $x_{N+1} = x_1$. If the directed edge ratios are

$$q_i = \frac{x_i}{x_{i+1}}, \qquad i = 1, \ldots, N, \tag{1}$$

then

$$\prod_{i=1}^{N} q_i = 1. \tag{2}$$

This follows by cancellation. For positive quantities, the equivalent log relation is

$$\sum_{i=1}^{N} \log q_i = 0. \tag{3}$$

These are algebraic identities. Their validity does not depend on a behavioral theory, a sampling design, or a causal mechanism.

This paper applies that distinction to a G20 country panel. The repository constructs a three-edge cycle from population, land area, and the Gini index. Since the ratios are calculated from the same country-year values, the product closes by construction. The empirical analysis is a separate descriptive panel regression of log Gini on log population density and log real GDP per capita, including country and year fixed effects.

The paper addresses four questions:

1. What does the cyclic closure identity establish in this application?
2. What is the coverage and descriptive profile of the saved G20 panel?
3. What associations are reported by the repository's two-way fixed-effects model?
4. What limitations constrain inference, interpretation, and replication?

The paper does not interpret closure as evidence of an economic relationship. Nor does it interpret the regression as causal. Country and year fixed effects remove some forms of heterogeneity, but do not by themselves resolve time-varying confounding, reverse causality, measurement error, or selective observation of inequality data.

2 Algebraic framework

2.1 Cycle closure

For nonzero x_i, the cycle of ratios is defined by

$$q_i = \frac{x_i}{x_{i+1}}, \qquad x_{N+1} = x_1. \tag{4}$$

Its product is

$$\prod_{i=1}^{N} q_i = \frac{x_1}{x_2}\frac{x_2}{x_3}\cdots\frac{x_N}{x_1} \tag{5}$$

$$= 1. \tag{6}$$

For positive levels, set $z_i = \log x_i$ and $r_i = \log q_i$. Then

$$r_i = z_i - z_{i+1}, \tag{7}$$

$$\sum_{i=1}^{N} r_i = 0. \tag{8}$$

In graph terms, the vector of log edge ratios is the incidence-matrix transform of the vector of node log levels. A common rescaling of all levels leaves the ratios unchanged, so ratios encode relative levels, not overall scale.

2.2 Population, area, and Gini

For country i in year t, let P_{it} be population, A_i land area, and G_{it} the Gini index. For positive values, the repository's cycle is

$$C_{it} = \left(\frac{P_{it}}{A_i}\right)\left(\frac{A_i}{G_{it}}\right)\left(\frac{G_{it}}{P_{it}}\right) = 1. \tag{9}$$

The first edge is population density, with units of persons per square kilometre. The other edges are algebraic contrasts and are not given independent economic interpretations here. The Gini index must be positive for the displayed ratios and log transformations to be defined.

All three quotients are computed from the same country-year data vector. Thus the product equals one algebraically, apart from floating-point error. The saved summary reports a maximum absolute log product of 2.220×10^{-16}, which is consistent with numerical rounding.

This is not a test of noisy closure. A noisy-cycle test would require separately measured or estimated log edge ratios and their joint measurement or sampling-error covariance. The panel data instead provide levels from which the edges are constructed. The cycle result consequently verifies the arithmetic implementation; it does not validate the data, establish construct validity, or test a theory.

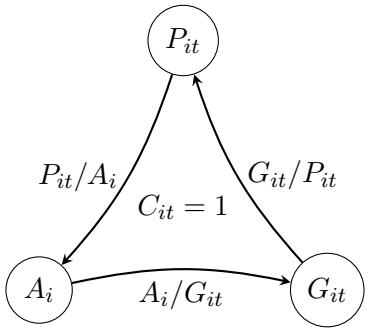


Figure 1: The population-area-Gini cycle.

(a) With common underlying levels, closure follows by cancellation.

3 Data and sample

3.1 Indicators and period

The repository's Python script requests World Bank Indicators for the period 2000–2023. It includes 19 sovereign country members in its country list and excludes the European Union and African Union aggregate rows. The saved panel contains 19 countries observed over 24 requested years, for 456 country-year rows.

Construct	Indicator code	Definition used in the script
Population	`SP.POP.TOTL`	Population, total
Land area	`AG.LND.TOTL.K2`	Land area, square kilometres
Gini index	`SI.POV.GINI`	World Bank Gini index
Real GDP per capita	`NY.GDP.PCAP.KD`	Constant 2015 US dollars

Table 1: World Bank indicators requested by the repository script.

Definitions and metadata should be archived with a final replication.

The saved outputs show complete observations for population, land area, and GDP per capita across all 456 rows. The Gini index is observed for 327 rows, or approximately 71.7% of the panel. Consequently, the panel regression uses 327 complete observations. Gini availability varies substantially by country, as shown in Table 2.

Country	Gini observations	Country	Gini observations
Argentina	23	Australia	9
Brazil	22	Canada	23
China	16	France	24
Germany	23	India	5
Indonesia	24	Italy	23
Japan	13	Korea, Rep.	11
Mexico	13	Russian Federation	24
Saudi Arabia	0	South Africa	6
Turkiye	22	United Kingdom	22
United States	24		

Table 2: Observed Gini values in the saved 2000–2023 country panel.

The unbalanced coverage affects the regression sample.

The table describes coverage, not the reason for missingness. Gini estimates are often based on household surveys and may differ in survey timing, income concepts, and coverage. The analysis does not impute or interpolate missing Gini values. Complete-case estimation may nevertheless be selective if observation availability is related to a country's economic conditions or statistical capacity.

3.2 Derived variables and descriptive statistics

Population density is calculated as

$$D_{it} = \frac{P_{it}}{A_i}. \tag{10}$$

The regression uses natural logarithms of Gini, population density, and real GDP per capita. These transformations require positive observations. Table 3 reports the saved output's descriptive statistics in levels.

Variable	N	Mean	Standard deviation	Minimum	Maximum
Population density	456	141.22	149.29	2.48	531.11
Gini index	327	38.21	7.43	23.00	65.00
GDP per capita	456	23,935.21	17,785.23	756.70	65,667.28

Table 3: Descriptive statistics from the saved repository output.

GDP per capita is in constant 2015 US dollars; density is persons per square kilometre.

The saved Gini values range from 23 to 65, with a mean of 38.21 and standard deviation of 7.43. These figures are on the scale as returned by the World Bank indicator. The scale, income concept, survey method, and metadata vintage should be confirmed and documented before publication. Population density and GDP per capita are available in all panel rows in the saved snapshot, but their distributions combine between-country differences and within-country time changes.

4 Econometric specification

The repository estimates a descriptive two-way fixed-effects model:

$$\log G_{it} = \alpha_i + \lambda_t + \beta_D \log D_{it} + \beta_Y \log Y_{it} + u_{it}, \tag{11}$$

where G_{it} is the Gini index, D_{it} is population density, Y_{it} is real GDP per capita, α_i denotes country fixed effects, λ_t denotes year fixed effects, and u_{it} is the error term.

The model is estimated by ordinary least squares with country and year indicators. Standard errors are clustered by country using the CRV1 covariance estimator as reported by the saved model output. In a log-log specification, β_D and β_Y describe conditional associations in log units. They can be read approximately as percentage associations for small changes, conditional on the included effects and the observed sample. They are not causal elasticities without a design and assumptions that justify causal interpretation.

Country fixed effects absorb time-invariant country characteristics, while year effects absorb shocks common to countries in a given year. These controls do not address time-varying confounders, reverse causality, measurement error in the Gini index or regressors, or potentially non-random survey coverage. Therefore, the specification is interpreted as descriptive.

5 Results

5.1 Panel model estimates

Table 4 summarizes the two regressors of substantive interest from the stored model output. The regression uses 327 observations from 18 countries over 24 years. Saudi Arabia contributes no Gini observations in the saved panel, so it is absent from the model sample.

Regressor	Coefficient	Clustered S. E.	p-value	95% C. I.
Log population density	0.0186	0.330	0.955	[−0.628, 0.665]
Log GDP per capita	0.0475	0.096	0.620	[−0.140, 0.235]

Table 4: Two-way fixed-effects regression estimates from the saved model output.

Country-clustered CRV1 standard errors are reported.

The estimated coefficient on log population density is positive but close to zero relative to its standard error. Its p-value is 0.955, and the reported 95% confidence interval includes both negative and positive values. The estimate therefore does not provide evidence of a statistically distinguishable conditional association in this specification.

The estimated coefficient on log GDP per capita is also positive and imprecise. Its p-value is 0.620, and its 95% confidence interval includes zero. This output likewise does not establish a statistically distinguishable conditional association in the specified sample and model.

These results should not be read as proof that density or income has no relationship with inequality. They indicate that this model, sample, and reported covariance estimator do not estimate either coefficient precisely enough to distinguish it from zero at conventional significance thresholds. A causal conclusion is not supported.

5.2 Model diagnostics and fit

The saved regression output reports $R^2 = 0.924$ and adjusted $R^2 = 0.913$. These statistics apply to the model including country and year indicators. They should not be attributed to the two substantive regressors alone: fixed effects can account for substantial variation, and a high overall R^2 is not evidence of causal validity or a well-specified mechanism.

The output also reports a condition number of 3.38×10^3, with a warning that strong multicollinearity or other numerical issues may be present. The condition number is a diagnostic flag, not a definitive diagnosis. The model's design matrix should be examined, particularly because density and GDP per capita can have substantial persistent between-country variation and the specification includes many indicator variables.

The reported Durbin–Watson statistic is 0.428. This statistic is not a complete panel serial-correlation test, especially for an unbalanced panel with country clustering and fixed effects. Nonetheless, the low value is a reason to investigate residual serial dependence rather than assume that the current covariance treatment settles all dependence concerns.

Model statistic	Saved output
Observations	327
Countries in regression	18
Years represented	24
R^2	0.924
Adjusted R^2	0.913
Model F-statistic	52.98
Reported model F-test p-value	2.08×10^{-11}
Condition number	3.38×10^3
Durbin–Watson statistic	0.428

Table 5: Selected model-level statistics from the saved output.

The overall model test includes fixed-effect terms and is not a test that the two substantive regressors are jointly causal.

The reported overall F-test concerns model terms collectively under the software's covariance procedure. Because the model includes many country and year indicators, its small reported p-value should not be interpreted as evidence that density or income has a causal effect. The coefficient-level uncertainty for the two variables of interest remains substantial.

5.3 Algebraic cycle diagnostic

The repository summary reports 456 panel rows, 327 Gini observations, and 327 complete observations for the logged regression. The maximum absolute log of the constructed cycle product is 2.220×10^{-16}. This numerical result is consistent with floating-point cancellation:

$$\log C_{it} = (\log P_{it} - \log A_i) + (\log A_i - \log G_{it}) + (\log G_{it} - \log P_{it}) \tag{12}$$

$$= 0. \tag{13}$$

It is not a statistical test, does not use a sampling distribution, and supplies no independent evidence about inequality or density.

6 Inference and limitations

6.1 Few country clusters

The regression includes 18 countries. Cluster-robust asymptotic inference is usually justified by a large number of independent clusters, and 18 is a limited number for conventional cluster asymptotics. The stored output applies a CRV1 correction, but that correction alone does not ensure accurate finite-sample confidence intervals or p-values. Further analysis should consider a suitable small-cluster procedure, such as a wild cluster bootstrap where its assumptions and implementation are appropriate, and should report the procedure and sensitivity of conclusions [3, 5].

6.2 Unbalanced and potentially selective Gini observations

The 327 observed Gini values are unevenly distributed across countries. Some countries have nearly complete coverage, while others have far fewer observations, and Saudi Arabia has none. If reporting years are related to economic events, survey availability, or statistical capacity, complete-case estimates may not represent the full 19-country panel. The regression's $N = 327$ does not by itself address this selection problem.

The paper does not impute missing Gini observations. Future work should report sensitivity to the observed sample and investigate the data-generating process for missingness. Any interpolation, imputation, or alternative survey series would change the estimand and should be justified and documented rather than treated as a routine repair.

6.3 Measurement and comparability

Cross-country Gini data can differ in underlying welfare concept, survey design, population coverage, and timing. The indicator's reported scale and metadata vintage should be verified. Land area is used as a denominator in density and is assumed comparable over the selected period; boundary or definition differences would affect cross-country comparisons. GDP per capita is reported in constant 2015 US dollars, but the exact source methodology and revision vintage should be archived for replication.

Measurement error in the Gini index may be substantial and need not be classical or independent across observations. Measurement error can attenuate or otherwise distort associations, while common survey or source errors can create dependence. The current model does not explicitly estimate a measurement-error process.

6.4 Model specification and interpretation

Country and year fixed effects do not eliminate all omitted-variable bias. Time-varying institutional, demographic, fiscal, labor-market, and distributional changes may affect both regressors and inequality. Reverse causality is also possible. Accordingly, the model is descriptive, not causal.

The relatively high condition number and low Durbin–Watson statistic warrant additional diagnostics. In a panel setting, researchers should inspect within-country residual dependence and assess the design matrix and regressor variation after absorbing fixed effects. A richer specification should be motivated by theory and data availability, not added mechanically to chase statistical significance.

7 Reproducibility and repository status

The analysis is implemented in the repository source file `src/empirical_g20_cqs.py`.

The saved outputs used here are `outputs/g20_world_bank_panel.csv`, `outputs/descriptive_statistics.csv`, `outputs/country_coverage.csv`, `outputs/data_and_cycle_summary.txt`, and `outputs/panel_model.txt`.

The current source contains numeric log transformations and a numeric coercion step before fitting the model.

The paper reports the stored outputs in the repository snapshot reviewed for this draft. Because the source retrieves data from a live API, future runs may differ if the World Bank revises historical observations or metadata. A formal replication archive should record:

1. the retrieval date and exact downloaded CSV;
2. Python and dependency versions;
3. the precise script revision and command-line arguments;
4. the complete model output and any diagnostics;
5. the indicator metadata and data definitions used.

The current paper does not claim that the results were independently rerun during drafting; numerical values are transcribed from the saved repository outputs.

8 Discussion

The G20 analysis illustrates two distinct roles for cyclic quotient systems and panel econometrics. The CQS identity establishes exact closure for the ratios constructed from common levels. Its observed numerical residual is approximately machine precision, as expected. That result is purely algebraic.

The panel regression addresses a different question: whether within-panel variation in density and real GDP per capita is associated with variation in the Gini index, conditional on country and year indicators. The point estimates are small and statistically imprecise under the reported covariance estimator. This does not demonstrate a zero substantive relationship, nor does the large overall model R^2 overturn the coefficient-level uncertainty. Sparse Gini coverage, few clusters, and diagnostics limit the strength of inference.

Welfare interpretation requires additional normative assumptions. A Gini index summarizes one aspect of a distribution and does not by itself determine mean income, poverty, mobility,

or a complete social welfare ordering [2, 6]. Similarly, an association between density and inequality would not establish a mechanism or policy effect. A causal study would need a defensible intervention, counterfactual, identification strategy, and assumptions appropriate to the research setting [1, 4].

9 Conclusion

The population-area-Gini cycle closes because the three ratios are constructed from one common vector of country-year levels. The saved maximum absolute log residual, 2.220×10^{-16}, is a floating-point consistency result, not an empirical discovery and not a test of noisy closure.

The separate two-way fixed-effects regression uses 327 observations from 18 countries and 24 years. Its estimated log-density coefficient is 0.0186 with standard error 0.330 and $p = 0.955$. Its estimated log-income coefficient is 0.0475 with standard error 0.096 and $p = 0.620$. Both reported confidence intervals include zero. These outputs do not provide evidence of precise conditional associations in this specification, and they support no causal claim.

Interpretation should remain cautious because Gini coverage is uneven, the number of clusters is limited, and the model diagnostics include a high condition number and low Durbin–Watson statistic. The appropriate contribution of the CQS framework here is to make the algebraic structure explicit and prevent an identity from being mistaken for evidence. Economic inference comes from the data, model, research design, and assumptions, not from cycle closure itself.

References

[1] J. D. Angrist and J.-S. Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton University Press, Princeton, 2009.

[2] A. B. Atkinson, "On the measurement of inequality," *Journal of Economic Theory*, vol. 2, no. 3, pp. 244–263, 1970.

[3] A. C. Cameron and D. L. Miller, "A practitioner's guide to cluster-robust inference," *Journal of Human Resources*, vol. 50, no. 2, pp. 317–372, 2015.

[4] G. W. Imbens and D. B. Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*, Cambridge University Press, Cambridge, 2015.

[5] J. G. MacKinnon and M. D. Webb, "Wild bootstrap inference for wildly different cluster sizes," *Journal of Applied Econometrics*, vol. 32, no. 2, pp. 233–254, 2017.

[6] A. Sen, *On Economic Inequality*, Clarendon Press, Oxford, 1973.

[7] J. M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, Cambridge, Massachusetts, 2010.

[8] World Bank, *World Development Indicators*, World Bank, Washington, DC. World Bank Indicators API and indicator metadata, `https://api.worldbank.org/`; accessed for the repository data snapshot.

Glossary

Algebraic closure
: The product-one identity generated by ratios arranged around a closed cycle.

Country fixed effect
: A country-specific intercept that absorbs time-invariant country characteristics in a panel specification.

Cycle residual
: The sum of observed oriented log-ratios around a cycle; it is zero for exact ratios derived from common levels.

Descriptive association
: A conditional relationship in observed data that is not interpreted as a causal effect.

Gini index
: A summary measure of inequality whose welfare concept, scale, survey basis, and population coverage should be documented.

Noisy-cycle test
: A statistical test of closure for separately measured edge ratios, requiring an error model and joint covariance.

Two-way fixed effects
: A panel specification containing unit and time intercepts.

Year fixed effect
: A year-specific intercept that absorbs shocks common to units during that year.

Cluster-robust standard error
: An uncertainty estimator that permits within-cluster dependence under suitable assumptions; few clusters require additional care.

The End